

Multi-Phase Aerodynamic Optimization of a Formula 1 Rear Wing Airfoil

Anthony Sanguinetti¹, Emily Parkerson¹, and Mariah Barsotti¹
Stanford University, AA222 / CS361 Engineering Design Optimization

This project presents a multi-phase aerodynamic optimization framework for the design of a Formula 1 rear wing section tailored to the Circuit de Monaco. Airfoil geometries were parameterized using a small set of geometric design variables and evaluated using the viscous-inviscid aerodynamic solver MSES. Bayesian optimization was used to efficiently explore the design space and identify improved designs based on a weighted objective function balancing downforce and drag. The optimization was performed in two phases, focusing on the main element geometry followed by the secondary element to maximize cornering performance. This phased approach reduced problem complexity while allowing the influence of each design component to be investigated independently. The resulting designs showed improved aerodynamic performance relative to the baseline configurations and produced trends consistent with a separate NSGA-II optimization study, demonstrating that Bayesian optimization provides an effective and computationally efficient approach for the aerodynamic design of multi-element Formula 1 rear wing sections.

I. Introduction

Formula 1 cars rely heavily on aerodynamic performance to maximize cornering grip while minimizing lap time. Of the aerodynamic components on the vehicle, the rear wing is one of the most important because it generates a significant portion of the total downforce while also contributing a large amount of drag [1]. Previous studies on Formula 1 rear wing optimization have shown how strongly rear wing geometry affects both aerodynamic loading and drag generation [2]. Unlike the front wing and undertray, the rear wing does not benefit significantly from ground effect, making it one of the least aerodynamically efficient devices on the car. Increasing rear wing loading improves tire grip and vehicle stability during braking and cornering, but it also reduces straight line speed because of the added drag. Rear wing design therefore becomes a balance between maximizing downforce and minimizing drag.

This tradeoff becomes especially important at tracks such as the Circuit de Monaco, where much of the lap is spent in low speed corners and braking zones rather than long straights. The Monaco circuit has one of the lowest average speeds on the Formula 1 calendar due to its tight layout and limited straight sections [3]. Under these conditions, maximizing downforce is generally more beneficial than minimizing drag, although excessive drag can still hurt acceleration and top speed in the few longer sections of the circuit.

The objective of this project is to develop a simulation-based optimization framework for a dual-element Formula 1 rear wing designed for the Circuit de Monaco. Parameterized airfoil geometries are analyzed using the viscous aerodynamic solver MSES [4], while Bayesian optimization methods [5] are used to search for improved designs. Rather than optimizing every design variable at once, the optimization process is separated into multiple phases that independently investigate primary element geometry, secondary element geometry, and secondary element angle of attack. The resulting airfoils are evaluated using a weighted scoring function based on downforce and drag performance.

II. Background and Related Work

A. Airfoil Optimization

The aerodynamic performance of an airfoil in a flow is dependent on the flow conditions and the airfoil characteristics. From these, the aerodynamic performance of the airfoil in freestream is suitably quantified by the coefficients of lift and drag, c_l and c_d , which are described for a 2D (sectional) airfoil as

$$c_l = \frac{2l}{\rho_\infty U_\infty^2 c}, \quad c_d = \frac{2d}{\rho_\infty U_\infty^2 c}, \quad (1)$$

¹Graduate Student, Aeronautical and Astronautical Engineering

where U_∞ and ρ_∞ are the freestream velocity and density, l and d are the lift and drag, and c is the chord length [6].

In general, as positive camber, angle of attack, or thickness increase, so do both lift and drag. The more frontal area is exposed to the flow, the higher the drag, and the larger the deflection of the air leaving the trailing edge from the freestream direction, the larger the lift.

In the case of motorsports, the goal is to produce negative lift, also known as downforce ($-c_l = d_f$), in order to maintain high grip on the track. In addition, any drag is considered a hindrance because it slows the vehicle, but this must be traded with downforce generation. Therefore, in order maximize vehicle performance, airfoils in the aerodynamic package of the body must be optimized to produce adequate downforce while minimizing drag [7]. Racecar aerodynamics, including downforce generation, downforce-drag tradeoffs, and wake interactions between bodies, are described in detail by Katz, and the slotted multi-element configuration used in modern Formula 1 rear wings extend this tradeoff by adding a second element downstream of the main element to delay separation at high lift.

B. Parameterization

Airfoil optimization requires a finite number of variables to represent the shape. There are two families of parameterization that dominate related literature. Direct geometric comparisons include maximum thickness and its location and maximum camber and its location, t_{max} , $x_{t,max}$, m , p . This method dates back to the NACA four and five digit series and remains in use for many design studies where physical interpretability is important. More flexible representations include the Class-Shape Transformation (CST) [8] and Bézier-curve methods, the latter of which is implemented in the Pymead toolkit [9]. Pymead is an open-source Python library and GUI designed for the generation, aerodynamic analysis, and shape optimization of parametric airfoils and airfoil systems [10]. It includes built in optimization of parametric airfoils and airfoil systems, and it uses both direct geometric parameters and Bézier control points as design variables. The choice of parameterization involves a tradeoff between dimensionality and expressiveness of the resulting optimization problem.

C. Aerodynamic Analysis Tools

Candidate airfoils must be quickly and reliably evaluated for aerodynamic performance for shape optimization to be feasible. Three classes of solver are commonly used. Panel methods coupled with integral boundary-layer equations, such as XFOIL [11], are fast and accurate, but they are limited to single-element sections at low to moderate Reynolds numbers. Reynolds-Averaged Navier-Stokes (RANS) solvers handle arbitrary geometries with full turbulence modeling, but these simulations typically cost minutes to hours and typically require the use of CFD software. Finally, MSES, a coupled viscous-inviscid Euler solver bridges the two former options.

Aerodynamic performance for each candidate airfoil geometry was evaluated using the MSES aerodynamic analysis package developed by Mark Drela [4]. MSES is a two-dimensional viscous aerodynamic solver that combines an inviscid Euler solution with an integral boundary layer formulation to model viscous effects such as skin friction, displacement thickness, and flow separation. This coupled viscous-inviscid approach provides reasonably accurate aerodynamic predictions while remaining computationally inexpensive compared to high-fidelity CFD methods.

Within this project, MSES was used to compute c_l and c_d for each candidate geometry generated during the optimization process, and was integrated directly into the optimization workflow through a custom Python interface. Although MSES is significantly faster than full RANS simulations, the analysis is restricted to two-dimensional flow, and convergence can become sensitive for highly cambered geometries, thin leading edges, or airfoils with geometric imperfections. Candidate geometries that failed to converge were assigned a penalty score rather than rejected outright. Despite these limitations, MSES provided a good balance between computational efficiency and aerodynamic fidelity for the present work.

D. Optimization Methods

Aerodynamic shape optimization approaches fall into a few categories. Gradient-based methods are historically used for high-fidelity shape design [12, 13]. However, these methods require the gradients to be smooth and easily found, and they struggle under noisy and discontinuous objectives. This noise and discontinuity can be produced by solver-convergence issues, making gradient methods non-ideal for integration with sensitive numerical solvers like MSES. Population-based algorithms, including genetic algorithms and NSGA-II have been applied to multi-element airfoils for high lift devices [14] and Formula SAE wings [15]. They handle non-smooth objectives but typically require a large magnitude of iterations to converge on an optimal solution. Bayesian optimization, which uses a Gaussian-process surrogate of the objective to guide sampling, has emerged as a competitive option for problems where each evaluation is expensive and gradients are unavailable or unreliable [16, 17].

1. Bayesian Optimization

From these multi-phase, multi-element, and high dimensionality, constrained problem characteristics, a problem-agnostic approach was determined necessary, and Bayesian Optimization is an effective algorithmic choice to fit the problem characteristics. Bayesian Optimization is a highly efficient, sequential strategy used to find the global maximum or minimum of “black-box” functions, and it is especially useful when gradients are unknown [18]. In general, Bayesian Optimization balances exploration and exploitation, using a surrogate model to approximate the true underlying function and an acquisition function to estimate the expected value of an unexplored point. Details of the Bayesian Optimization framework are given in Section III.C.

2. Non-Dominated Sorting Genetic Algorithms

Since this project was originally inspired by the Pymead aerodynamic optimization framework, the evolutionary optimization techniques employed by Pymead are briefly reviewed for comparison and validation purposes. Pymead utilizes a Unified Non-Dominated Sorting Genetic Algorithm (UNSGA-III) implementation from the pymoo optimization library [19]. This class of algorithms performs well for higher-dimensional optimization problems and is particularly effective for problems involving competing objective functions.

Non-dominated sorting genetic algorithms belong to a broader family of population-based evolutionary optimization methods. These algorithms evolve populations of candidate solutions using biologically inspired operations such as selection, crossover, and mutation. Evolutionary optimization methods are particularly effective for nonlinear, nonconvex, and multimodal optimization problems because they perform probabilistic exploration of the design space rather than relying on deterministic gradient-based search methods [20].

To validate the custom Bayesian Optimization framework developed in this project, a Pymead inspired evolutionary optimization workflow based on the Non-Dominated Sorting Genetic Algorithm II (NSGA-II) was implemented using pymoo [21]. This approach was chosen instead of Pymead’s default UNSGA-III due to the reduced number of design variables and objectives considered in the present work. The primary focus of this project is the development and analysis of the Bayesian Optimization methodology, while the NSGA-II implementation is primarily for validation, comparison, and evaluating the robustness and consistency of the resulting aerodynamic optimization trends.

III. Methodology

A. Geometric Parameterization and Design Variables

The rear wing optimization was separated into two phases. Phase 1 focused on optimizing the shape of the primary element at a fixed angle of attack. Phase 2 then optimized the shape of the secondary element while maintaining a fixed relative placement with respect to the primary element. Separating the optimization into phases reduced the dimensionality of the design space while still allowing the effects of the primary and secondary element shapes to be evaluated independently. For Phases 1 and 2, each airfoil geometry was parameterized using direct parameterization:

Table 1 Airfoil Design Variables Used for Geometry Parameterization

Variable	Definition
m	Maximum camber of the airfoil, normalized by chord length.
p	Chordwise location of maximum camber, normalized by chord length.
$t_{\max}$	Maximum thickness of the airfoil, normalized by chord length.
$x_{t,\max}$	Chordwise location of maximum thickness, normalized by chord length.

To begin the geometric parameterization, the seed airfoil coordinates were first normalized by chord length and rotated to remove the seed angle of attack. The coordinates were then split into upper and lower surfaces, resampled, and used to compute the seed camber and thickness distributions using Eq. (2) and Eq. (3).

$$y_c(x) = \frac{y_u(x) + y_l(x)}{2} \quad (2)$$

$$t(x) = y_u(x) - y_l(x) \quad (3)$$

Cosine spacing was used for rebuilding the geometry from the thickness and camber distributions:

$$x_i = \frac{c}{2} (1 - \cos \theta_i), \quad i = 0, \dots, N-1 \quad (4)$$

where c represents the chord length of the airfoil and $\theta = [0, \pi]$. By using cosine spacing, the points along the airfoil are more highly concentrated near the leading and trailing edges, which is important because small oscillations or irregularities in these regions can cause issues with MSES convergence.

The new thickness distribution was generated by morphing the seed thickness distribution. First, the location of the maximum seed thickness was identified. A coordinate mapping was then constructed between the seed peak location and desired peak location using five control points.

The Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) interpolation method was used to fit a shape-preserving, smooth, continuous curve to the seed coordinate mapping, which results in the new coordinate mapping with the desired thickness coordinates. Once the coordinates were remapped, the amplitudes of the points were rescaled by a ratio of the new maximum thickness to the maximum seed thickness.

Rather than directly perturbing the seed airfoil geometry, a new camber distribution was generated using the analytical camber line formulation of the NACA 4-digit series [22]. This parameterization provides direct control over the magnitude and location of maximum camber while maintaining a smooth and realistic camber distribution:

$$y_c(x) = \begin{cases} \frac{m}{p^2} (2px - x^2), & 0 \leq x < p \\ \frac{m}{(1-p)^2} [(1-2p) + 2px - x^2], & p \leq x \leq 1 \end{cases} \quad (5)$$

The upper and lower surfaces were reconstructed from the camber and thickness distributions. The local camber angle was computed as

$$\theta(x) = \tan^{-1} \left(\frac{dy_c}{dx} \right) \quad (6)$$

Using Eq. (7) and Eq. (8), the thickness is applied normal to the camber.

$$x_u = x - \frac{t}{2} \sin \theta, \quad y_u = y_c + \frac{t}{2} \cos \theta \quad (7)$$

$$x_l = x + \frac{t}{2} \sin \theta, \quad y_l = y_c - \frac{t}{2} \cos \theta \quad (8)$$

However, to improve robustness near the leading and trailing edges, the implementation blends the local camber angle toward zero in these regions. This causes the thickness to be applied more vertically near sensitive regions of the airfoil, reducing geometric oscillations.

For Phase 2, the same geometry generation process was used for the secondary element. After reconstruction, the secondary element was scaled and translated:

$$\begin{bmatrix} x_{\text{scaled}} \\ y_{\text{scaled}} \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} + s \left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \right) \quad (9)$$

$$x' = x + \Delta x, \quad y' = y + \Delta y \quad (10)$$

The angle of attack corresponding to cornering conditions was fixed to match the seed airfoil configuration.

B. Objective Function

Within the optimization framework, the objective is to maximize the performance metric $f(\mathbf{x})$ subject to the prescribed design constraints:

$$\begin{aligned} \max_{\mathbf{x}} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \\ & h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p. \end{aligned} \quad (11)$$

1. Scoring Function

The desirability of an airfoil based on its aerodynamic performance can be quantified via a scoring function with a relative weighting between downforce and drag. The scoring function is given as follows:

$$f = W_{df} * c_{df} + W_{drag} * c_d \quad (12)$$

where W is the weighting both aerodynamic metrics. To determine weightings, Circuit de Monaco, seen in Figure 1, was decomposed into segments where straight driving, braking and acceleration, and cornering are expected by a typical driver at the Monaco Grand Prix.

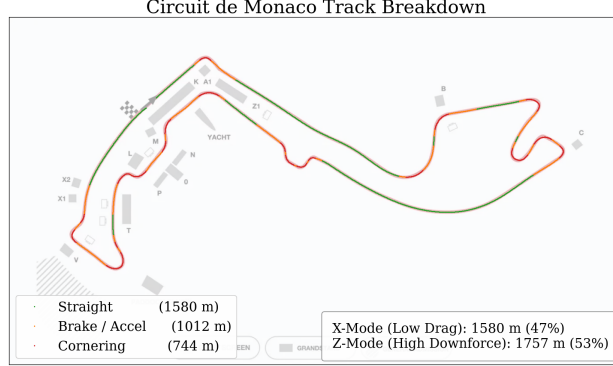


Fig. 1 Circuit de Monaco decomposed into low drag and high downforce segments.

An F1 vehicle has the ability to transition its rear wing from X-Mode to Z-Mode. X-Mode is a low drag mode where the second element is re-rotated to an angle of attack more aligned with the flow, and this mode is used when speed is most useful (e.g. straights). Z-Mode is a high downforce configuration where the secondary element rotates to a higher (negative) angle of attack, thus producing more downforce and drag. This is used when grip is extremely important (e.g. cornering, braking, accelerating).

Table 2 Aerodynamic Metric Weights by Optimization Phase

Phase	W_{df}	W_d	Primary Objective
1	0.53	-0.47	Balanced downforce and drag performance
2	0.65	-0.35	Prioritize downforce for cornering conditions
3	0.35	-0.65	Heavily penalize drag for straight-line performance

Phase 1 (P1) only optimizes the main element, which is at a fixed angle of attack. Thus it must be optimal for the entire track in one configuration, so the weightings are proportional to the circuit layout to have the most beneficial shape for both conditions simultaneously.

In Phase 2 (P2), the main element continues to be fixed, and the secondary element is added to the configuration in Z-Mode. Because the track composition prioritizes high downforce, the geometric properties of the airfoil itself should be catered to best Z-Mode performance. Thus, downforce is prioritized in the weighting scheme.

2. Constraints

The prescribed constraints, $\mathbf{g}(\mathbf{x})$, for this project include the following:

$$t_{\max} > m, \quad (13)$$

$$x_{t,\max} < p, \quad (14)$$

$$t_{\max,\text{lower}} \leq t_{\max} \leq t_{\max,\text{upper}}, \quad (15)$$

$$m_{\text{lower}} \leq m \leq m_{\text{upper}}, \quad (16)$$

$$x_{t,\text{lower}} \leq x_{t,\max} \leq x_{t,\text{upper}}, \quad (17)$$

$$p_{\text{lower}} \leq p \leq p_{\text{upper}}. \quad (18)$$

Equation 13 prevents the camber line from exceeding the half-thickness, which would force the upper and lower surfaces to intersect near the leading edge. Equation 14 requires the maximum-thickness location to lie forward of the

maximum-camber location, which is consistent with low Reynolds number airfoil convention. Each design variable's upper and lower bound is specified based on the optimization phase bound given in Table 5. These geometric feasibility constraints are imposed on every candidate design to ensure that the airfoil shape is well-formed and the resulting geometry can be analyzed by MSES, which is highly sensitive to off-nominal airfoil designs. If any of the constraints are violated, the airfoil is deemed invalid and not run through aerodynamic analysis.

These constraints are enforced at two points in the optimization loop. During initial training data generation, any infeasible design is discarded and a new design is sampled. During the BO loop, the expected improvement acquisition function is maximized over the feasible region of the normalized design space, so the proposed point is feasible by construction. Therefore, no computation is wasted on aerodynamic analysis of infeasible designs.

3. Invalid-Result Handling

A proposed design may satisfy the constraints detailed above and still fail to converge in MSES due to the sensitive nature of the coupled inviscid-viscous solver. Often this happens because the geometry produces flow features the solver is unable to resolve including stalled boundary layers, wakes, or separation bubbles that fail to reattach.

Any evaluation that cannot converge under the internal iteration and time limits for MSES and MSET is assigned a score $f = -1$. This is significantly worse than any plausible converged score in the weighted scoring function. The GP then fits to data including these penalty values, and the acquisition function is deterred from proposing candidates which may return similar errors.

C. Bayesian Optimization Workflow

The Bayesian Optimization (BO) loop combines probabilistic surrogate modeling of the objective function with an acquisition function that balances exploitation of high performing regions with exploration of uncertain regions within the design space. The objective $f(\mathbf{x})$ is the score defined in Eq. 12. The design vector is $\mathbf{x} = [m, m \text{ location}, t_{max}, t_{max} \text{ location}]$.

1. Gaussian Process Surrogate Model

The objective function is modeled as a Gaussian process (GP). This is a distribution over functions, and it is characterized by a mean function $m(\mathbf{x})$ and a covariance (kernel) function $k(\mathbf{x}, \mathbf{x}')$ [16]. Given a training set $X = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(|X|)}\}$ with observed scores $\mathbf{y}$, the joint prior is

$$\begin{bmatrix} y_1 \\ \vdots \\ y_{|X|} \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} m(\mathbf{x}^{(1)}) \\ \vdots \\ m(\mathbf{x}^{(|X|)}) \end{bmatrix}, \mathbf{K} \right), \quad K_{ij} = k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}). \quad (19)$$

A squared exponential kernel is employed,

$$k(\mathbf{x}, \mathbf{x}') = \exp \left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2} \right), \quad (20)$$

with characteristic length scale ℓ , which results in smoother functions as the value increases. As the value of ℓ increases, more correlation is applied between distant points in the design space.

In the case of our project, each design variable has different bounds, so the design variables are each normalized before the kernel is evaluated. The length scale is re-fit at each BO iteration by maximizing the GP's marginal likelihood on the current training data.

Given the training set and fitted GP, the posterior predictive distribution at a new test point $\mathbf{x}^*$ is Gaussian with mean $\mu(\mathbf{x}^*)$ and variance $\sigma^2(\mathbf{x}^*)$:

$$\mu(\mathbf{x}^*) = \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{y}, \quad \sigma^2(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{k}_*, \quad (21)$$

where $\mathbf{k}_* = [k(\mathbf{x}^*, \mathbf{x}^{(i)})]_{i=1}^{|X|}$.

2. Acquisition Function

To select the next design point to evaluate, the Expected Improvement (EI) acquisition function is used. The following derivation in [16] is expressed in maximization form here. With the best observed score y_{best} , the improvement at a candidate point $\mathbf{x}$ is the random variable

$$I(\mathbf{x}) = \max(f(\mathbf{x}) - y_{\text{best}}, 0), \quad (22)$$

and its expectation under the GP posterior has the closed form

$$\text{EI}(\mathbf{x}) = (\mu(\mathbf{x}) - y_{\text{best}}) \Phi(z) + \sigma(\mathbf{x}) \phi(z), \quad z = \frac{\mu(\mathbf{x}) - y_{\text{best}}}{\sigma(\mathbf{x})}, \quad (23)$$

where Φ and ϕ are the standard normal cumulative distribution function (CDF) and probability density function (PDF). The PDF is the height of the bell curve at z for the standard normal random variable Z , and it quantifies the relative likelihood of the value z . The CDF is the area under the bell curve from $-\infty$ to z , and it quantifies the probability that the random variable is $\leq z$.

EI is high wherever the predicted mean exceeds the best observed score and where uncertainty is large, encouraging exploitation and exploration, respectively. At each iteration, the next point $\mathbf{x}^*$ is chosen by maximizing EI over the normalized design space subject to the geometric constraints, and then it is de-normalized before evaluation.

3. Algorithm

The full BO procedure used in this work is summarized below:

- 1) Evaluate the objective at the seed design $\mathbf{x}_{\text{seed}}$ to obtain $f(\mathbf{x}_{\text{seed}})$, and append the pair $(\mathbf{x}_{\text{seed}}, f(\mathbf{x}_{\text{seed}}))$ to the training set.
- 2) Perturb $\mathbf{x}_{\text{seed}}$ within the design bounds (rejecting samples that violate the feasibility constraints) to obtain $\mathbf{x}_{\text{train}}$, evaluate for f_{train} , and append all resulting pairs to the training set.
- 3) Normalize the training inputs to $[0, 1]^d$ and fit a Gaussian process to the $(\mathbf{x}_{\text{train}}, f_{\text{train}})$ pairs, re-fitting the squared-exponential length scale ℓ by maximum marginal likelihood.
- 4) Use the GP posterior together with the expected improvement acquisition function (Equation 23) to propose a new design point $\mathbf{x}^*$.
- 5) Denormalize $\mathbf{x}^*$ and evaluate the objective at it to obtain $\hat{y} = f(\mathbf{x}^*)$. If MSES fails to converge, $\hat{y}$ is set to a penalty value of -1 .
- 6) Append $(\mathbf{x}^*, \hat{y})$ to the training set.
- 7) Repeat from step 3) until $N_{\text{conv}} = 400$ converged evaluations have been collected.
- 8) Return the best design point from 400 successful iterations.

D. Validation of Optimization

To validate the Bayesian Optimization framework developed in this project, the resulting airfoil geometries and score histories are compared against those produced using an alternative evolutionary optimization approach. As previously discussed, the Pymead aerodynamic optimization framework uses a UNSGA-III implementation through the pymoo optimization library. Inspired by this workflow, a simplified NSGA-II optimization framework was implemented for comparison and validation purposes.

Although NSGA-II is traditionally formulated as a multi-objective optimization algorithm, the present work utilizes a single scalar objective function based on the aerodynamic scoring function described previously, subject to geometric and aerodynamic constraints. Rather than generating a true Pareto front between competing objectives, the NSGA-II method effectively ranks candidate solutions according to higher or lower score values while still benefiting from the exploratory behavior of evolutionary optimization methods.

To enable direct comparison between optimization methodologies, the Pymead-inspired optimization workflow was modified to interface directly with the same geometric parameterization using the Bayesian Optimization method. The original Pymead framework primarily utilizes Bezier curve geometric parameterization and higher-dimensional variables, which would likely produce non-comparable optimization trends relative to the four-variable parameterization used in this project. Therefore, the NSGA-II implementation was adapted to use the same airfoil parameterization, geometric bounds and constraints, MSES aerodynamic evaluation, and aerodynamic scoring function as the Bayesian Optimization method. This ensured that differences in optimization behavior reflected the optimization algorithms themselves. The implementation architecture and workflow were conceptually inspired by the shape optimization and airfoil matching files in the Pymead code base [23].

After several preliminary trial runs, a termination criterion of 400 converged aerodynamic evaluations was selected to ensure reasonable convergence. A population size of 20 candidates was used. Since MSES does not converge successfully for every candidate geometry, the optimization algorithm was modified to continue iterating until 400 converged geometries were obtained. A generation limit of 600 was used only as a safety factor to prevent indefinite

runtime in the event of excessive solver divergence.

IV. Experimental Setup

A. Baseline Geometry and Flow Conditions

The baseline rear wing geometry used throughout this project was obtained from a publicly available Formula 1 rear wing CAD tutorial [24]. Airfoil coordinates for both the first and second elements of the inboard and outboard sections were extracted using WebPlotDigitizer and post-processed to remove duplicate points, normalize the chord length, and smooth minor geometric irregularities before aerodynamic analysis. These geometries served as the seed airfoils for all subsequent optimization phases. The geometric parameters for these airfoils can be seen in the following section.

The flow environment across the span of the rear wing is not uniform, thus separate operating conditions were selected for the inboard and outboard sections. The incoming flow to the inboard section is in the turbulent wake of the chassis, halo, and driver, thus it has a lower velocity and nonzero angle of attack. Meanwhile, the incoming flow to the outboard section is more freestream, and therefore has a higher velocity and zero angle of attack. The selected angles of attack of the incoming flow were estimated from publicly available stream trace and cross-sectional CFD visualizations of modern Formula 1 rear wing flow fields [25]. Vehicle velocities were estimated using Monaco race conditions reported in [3]. Reynolds number calculations were performed using a characteristic chord length of 0.415 m, which was selected based on FIA rear wing requirements discussed in previous Formula 1 aerodynamic studies [2]. Standard sea level atmospheric properties were used for density and dynamic viscosity.

The simulations were performed in MSES under low-Mach conditions to reflect what is seen at Monaco. Since all operating points remained below Mach 0.13, compressibility effects were relatively small. Reynolds numbers were on the order of 10^6 , which is consistent with low speed motorsport simulations. The resulting flow conditions are summarized in Table 3.

Table 3 Baseline Flow Conditions

Parameter	Inboard Section	Outboard Section
Velocity (km/h)	115	150
Mach Number	0.094	0.122
Angle of Attack (deg)	-2.5	0.0
Chord Length (m)	0.415	0.415
Freestream Density (kg/m ³)	1.225	1.225
Dynamic Viscosity (kg/(m·s))	1.7894×10^{-5}	1.7894×10^{-5}
Reynolds Number	9.08×10^5	1.18×10^6

B. Optimization Cases

The optimization process was decomposed into two phases. Phase 1 (P1) optimized main and secondary inboard (IB) and outboard (OB) elements. These were passed into P2 multi-element configurations. Both phases retained the original angles of attack, α . These optimization scopes are summarized in Table 4.

Table 4 Optimization cases and design variables.

Case	Configuration	Scope	Design Variables	Metrics
P1 IB	Single element	Inboard main element airfoil	$t_{max}, x_{t,max}, m, p$	f, c_l, c_d
P1 OB	Single element	Outboard main element airfoil	$t_{max}, x_{t,max}, m, p$	f, c_l, c_d
P2 IB	Multi-element	Inboard secondary element airfoil	$t_{max}, x_{t,max}, m, p$	f, c_l, c_d
P2 OB	Multi-element	Outboard secondary element airfoil	$t_{max}, x_{t,max}, m, p$	f, c_l, c_d

For each optimization case, the design variables were constrained by the bounds given in Table 5.

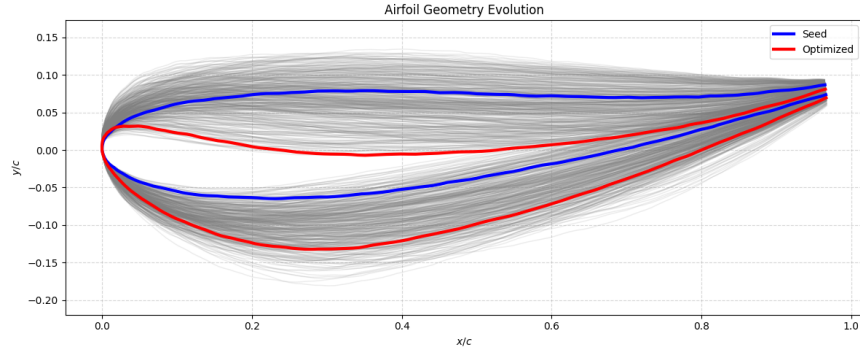
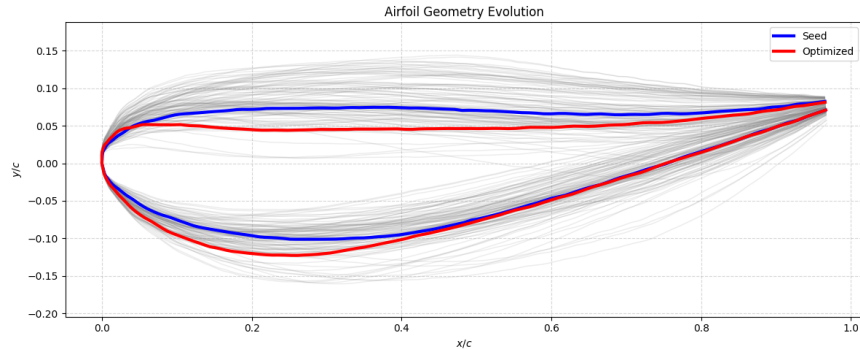
Table 5 Design variable bounds for each optimization case.

Case	$t_{\max}$	$x_{t,\max}$	m	p
P1 IB	[0.17, 0.22]	[0.15, 0.35]	[0.00, 0.15]	[0.30, 0.70]
P1 OB	[0.14, 0.22]	[0.15, 0.35]	[0.00, 0.10]	[0.30, 0.70]
P2 IB	[0.10, 0.18]	[0.20, 0.40]	[0.02, 0.08]	[0.50, 0.75]
P2 OB	[0.10, 0.18]	[0.20, 0.40]	[0.02, 0.08]	[0.50, 0.75]

V. Results

A. Phase 1: First Element Optimization

Figure 2 compares the seed and optimized airfoil geometries for both operating conditions. For the outboard case, the optimizer converged toward a significantly more cambered geometry while maintaining nearly the same thickness distribution. The increase in camber produced substantially greater aerodynamic loading. For the inboard case, the optimizer produced more moderate geometric changes, consisting primarily of increased camber and forward movement of both the camber and thickness locations. This behavior is consistent with the inboard flow conditions and will be explained in a subsequent paragraph.

**(a) OB airfoil evolution****(b) IB airfoil evolution****Fig. 2 Phase 1 airfoil evolution for the outboard and inboard operating conditions.**

The aerodynamic consequences of these geometric changes are illustrated through the pressure coefficient distributions, which are provided in Appendix A. For both operating conditions, the optimized airfoils develop stronger suction peaks and maintain lower pressure coefficients over larger portions of the lower surface compared to the seed geometries. These changes are consistent with the increased camber observed in the optimized airfoils and explain the resulting increase in downforce.

Figure 3 summarizes the explored aerodynamic design spaces, while Figure 4 shows the design variable combinations explored throughout the optimization process. Additional optimization histories and pressure coefficient comparisons

are provided in Appendix A.

The outboard optimization exhibited a clear lift-to-drag tradeoff and quickly identified high performing regions of the design space. Most of the performance improvement occurred within the first 100 valid evaluations, after which the optimizer primarily refined solutions near the best performing region. The optimizer explored a broad range of camber and thickness combinations and consistently converged toward highly cambered geometries that generated significantly larger downforce.

The inboard optimization displayed a similar trend but produced a smaller spread of aerodynamic performance. Although approximately 400 optimization iterations were attempted, only a little over 100 candidate geometries produced converged MSES solutions. The reduced number of valid evaluations was primarily caused by MSES convergence failures. The incoming flow to the inboard section is at an angle of attack of -2.5° , which produces a more challenging flow field than the outboard condition. As the optimizer increased camber in an attempt to improve aerodynamic performance, the resulting adverse pressure gradients became increasingly susceptible to flow separation and numerical convergence difficulties. As a result, many candidate geometries could not be successfully evaluated, reducing the effective size of the feasible design space available to the optimizer. This behavior partially explains why the inboard optimization achieved smaller performance improvements than the outboard optimization despite the same number of optimization iterations.

The optimized geometries and aerodynamic coefficients for both operating conditions are summarized in Table 6.

Table 6 Phase 1 Optimization Results

Parameter	OB Seed	OB Optimized	IB Seed	IB Optimized
c_l	-0.8711	-1.4541	-1.2128	-1.3612
c_d	0.01284	0.01739	0.01740	0.01851
Maximum Camber	0.0254	0.0977	0.0438	0.0621
Camber Location	0.6500	0.3616	0.4500	0.3000
Maximum Thickness	0.1421	0.1420	0.1767	0.1708
Thickness Location	0.2919	0.1786	0.2400	0.1500
Optimization Score	0 (baseline)	0.2483	0 (baseline)	0.0949

For the outboard operating condition, the optimization increased the lift coefficient from $c_l = -0.8711$ to $c_l = -1.4541$, corresponding to an increase in downforce of approximately 67%. This improvement was achieved primarily through a large increase in maximum camber, from 2.54% chord to 9.77% chord, while maintaining nearly the same thickness distribution. The optimizer also significantly moved the camber location forward, from 65% chord to approximately 36% chord, creating a more front-loaded pressure distribution.

For the inboard operating condition, the lift coefficient improved from $c_l = -1.2128$ to $c_l = -1.3612$, corresponding to an increase in downforce of approximately 12%. The optimizer increased maximum camber from 4.38% chord to 6.21% chord while slightly reducing thickness and shifting both the camber and thickness locations forward along the chord. Compared to the outboard case, the resulting geometric changes were considerably smaller and produced a more modest aerodynamic benefit.

Overall, the results indicate that airfoil camber was the dominant design variable controlling aerodynamic performance for both operating conditions. However, the outboard section exhibited far greater sensitivity to geometric modifications and therefore produced significantly larger performance gains. The optimizer consistently favored moving the camber location forward, suggesting that front-loaded loading distributions were beneficial under both operating conditions.

B. Phase 2: Second Element Optimization

Phase 2 focused on optimizing the second element of the rear wing while holding the Phase 1 optimized main elements fixed. Unlike Phase 1, the objective of this phase was to improve the aerodynamic performance of the complete two element configuration by modifying only the second geometry.

Figure 5 compares the seed and optimized second element geometries for both operating conditions. The Phase 2 seed geometry was identical for the outboard and inboard studies. However, because the second element operates within the wake and pressure field generated by the Phase 1 optimized main element, the resulting aerodynamic performance differs between the two operating conditions.

For both cases, the optimizer converged toward thinner and more highly cambered second element geometries. The optimized outboard second element exhibits a substantial increase in camber along with a reduction in thickness, producing stronger loading while maintaining favorable interaction with the upstream element. A similar trend is observed for the inboard case, although the geometric changes are somewhat less pronounced. These modifications increase the

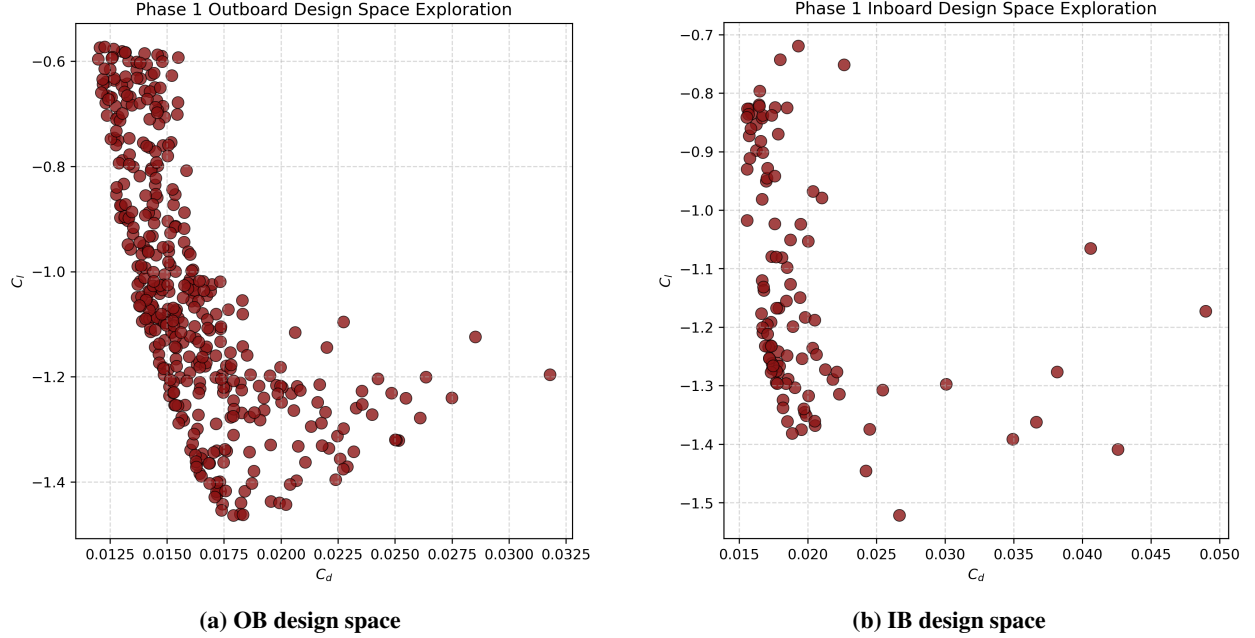


Fig. 3 Explored lift-to-drag design spaces for the Phase 1 optimization studies.

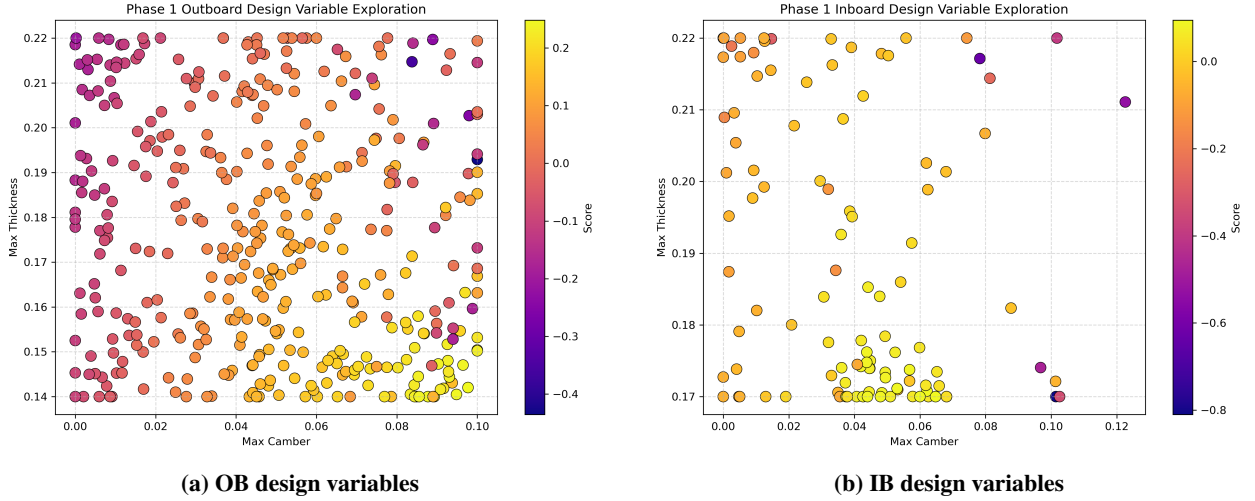


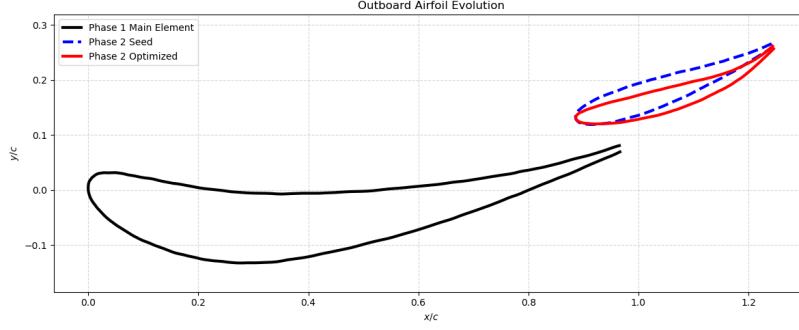
Fig. 4 Explored design variable combinations during the Phase 1 optimization studies. Color indicates the corresponding optimization score.

suction developed on the second element and improve the overall loading distribution across the multi-element airfoil system.

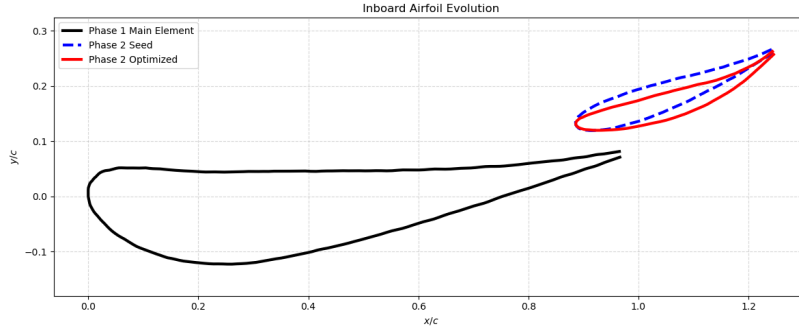
The aerodynamic consequences of these geometric changes are illustrated through the pressure coefficient distributions provided in Appendix A. For both operating conditions, the optimized second element develops stronger suction peaks and maintains lower pressure coefficients compared to the seed geometry. The increased suction is particularly evident on the lower surface, indicating that the optimizer successfully increased the loading carried by the second element while preserving favorable interaction between the two airfoil elements.

The Bayesian optimization framework evaluated approximately 40 valid geometries for the outboard study and approximately 80 valid geometries for the inboard study. The reduced number of valid evaluations relative to Phase 1 reflects the increased computational cost and convergence challenges associated with multi-element MSES analyses. Figure 6 summarizes the explored aerodynamic design spaces, while Figure 7 shows the design variable combinations explored throughout the optimization process. Additional optimization histories and pressure coefficient comparisons are provided in Appendix A.

The outboard optimization quickly identified high performing regions of the design space. The best score improved



(a) Outboard second element evolution



(b) Inboard second element evolution

Fig. 5 Phase 2 second element evolution for the outboard and inboard operating conditions. The main element remains fixed at the Phase 1 optimized geometry.

significantly within the first few valid evaluations before gradually converging toward the final optimum. The design space exploration reveals a clear tradeoff between drag and downforce, with the highest performing solutions corresponding to geometries with increased camber and reduced thickness. After approximately 30 valid evaluations, only small gains were obtained, indicating that the optimizer had effectively converged to a near optimal region of the design space.

The inboard optimization exhibited similar behavior but explored a broader range of candidate geometries. The convergence history shows several distinct improvements throughout the optimization process, indicating that the optimizer continued discovering higher performing regions of the design space well beyond the initial iterations. The design variable exploration suggests that increased camber was again the dominant factor driving aerodynamic performance. Higher scoring solutions consistently clustered near the upper end of the allowable camber range, while thinner second element geometries generally produced better overall performance.

The optimized geometries and aerodynamic coefficients for both operating conditions are summarized in Table 7.

Table 7 Phase 2 Optimization Results

Parameter	OB Seed	OB Optimized	IB Seed	IB Optimized
c_l	-2.7616	-3.1403	-2.8906	-3.2193
c_d	0.02577	0.02615	0.03391	0.03641
Maximum Camber	0.0254	0.0789	0.0254	0.0789
Camber Location	0.6500	0.6736	0.6500	0.6437
Maximum Thickness	0.1421	0.1158	0.1421	0.1254
Thickness Location	0.2919	0.3578	0.2919	0.3936
Optimization Score	0	0.1371	0	0.1137

For the outboard operating condition, the optimization increased the lift coefficient from $c_l = -2.7616$ to $c_l = -3.1403$, corresponding to an increase in downforce of approximately 13.7%. This improvement was achieved through a substantial increase in second element camber, from 2.54% chord to 7.89% chord, together with a reduction in thickness from 14.21% chord to 11.58% chord. The optimizer also moved the thickness location aft, producing a

geometry that generated stronger loading while maintaining reasonable drag characteristics. Despite the increased downforce, the drag coefficient only increased slightly from $c_d = 0.02577$ to $c_d = 0.02615$.

For the inboard operating condition, the lift coefficient improved from $c_l = -2.8906$ to $c_l = -3.2193$, corresponding to an increase in downforce of approximately 11.4%. Similar to the outboard case, the optimizer increased camber and reduced thickness while making more moderate adjustments to the camber and thickness locations. The drag coefficient increased from $c_d = 0.03391$ to $c_d = 0.03641$, reflecting the stronger loading generated by the optimized second element. Nevertheless, the resulting increase in downforce outweighed the drag penalty and produced a higher overall optimization score.

Overall, the Phase 2 results demonstrate that significant aerodynamic improvements can be achieved through second element optimization even after the main element has already been optimized. For both operating conditions, the optimizer consistently favored highly cambered and relatively thin second element geometries. The resulting designs generated stronger suction on the second element and increased overall downforce while maintaining reasonable drag levels. These trends suggest that camber was the dominant parameter governing second element performance and that increased loading on the second element remained beneficial under both operating conditions.

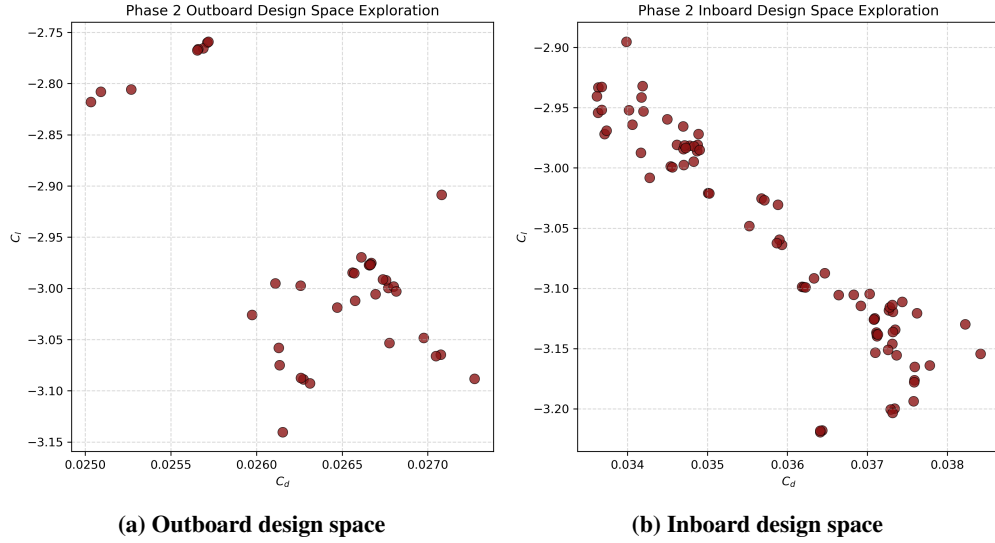


Fig. 6 Explored lift-to-drag design spaces for the Phase 2 optimization studies.

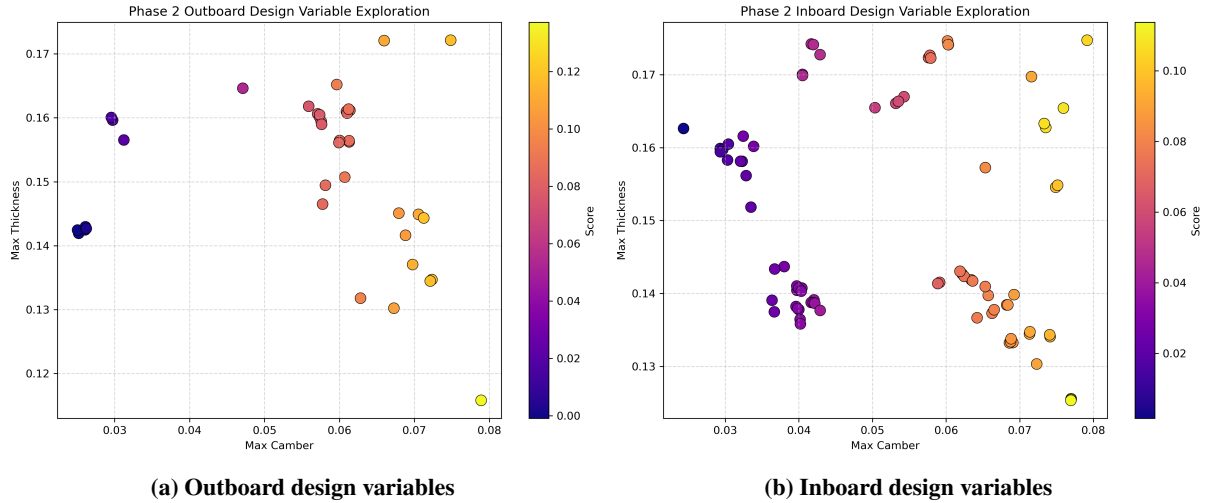


Fig. 7 Explored design variable combinations during the Phase 2 optimization studies. Color indicates the corresponding optimization score.

VI. Discussion

A. Optimized Airfoil vs. Common High Lift/Low Reynolds Number Airfoils

The optimized airfoil geometries exhibit several characteristics commonly observed in high lift, low Reynolds number airfoils contained in the UTUC Airfoil Database, including the Selig S1210, Selig S1223, and Church Hollinger CH 10-48-13 airfoils [26]. The Selig airfoils are commonly used within the Formula SAE community due to their favorable performance, providing a useful benchmark for comparison with the optimized designs generated in this project.

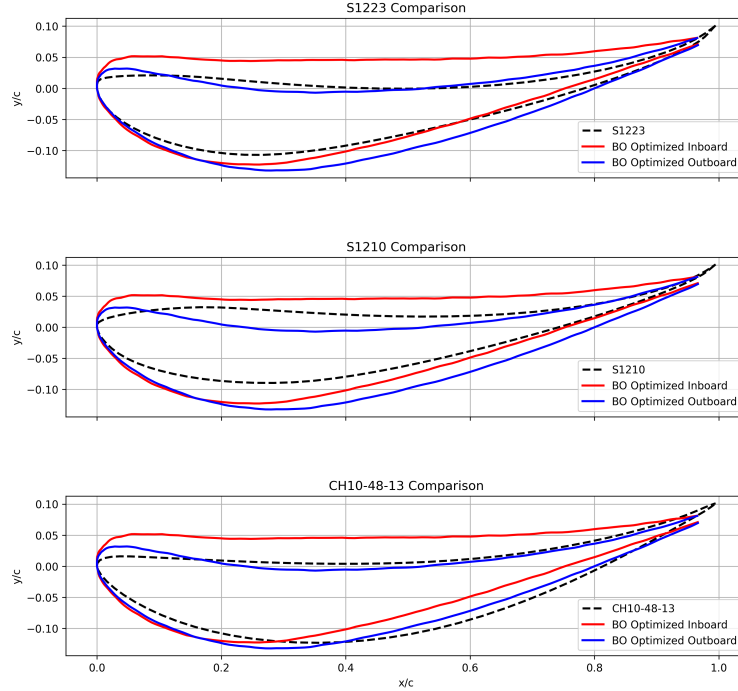


Fig. 8 Comparison of Phase 1 optimized airfoils with established high lift low Reynolds number airfoils.

Figure 8 compares the optimized inboard and outboard geometries against the selected reference airfoils. Although the optimized airfoils do not exactly reproduce any existing design, they exhibit several common features associated with high lift low Reynolds number airfoils, including increased camber. This agreement suggests that the optimization framework converged toward aerodynamic features known to be effective under the present flow conditions.

Notable differences are observed in the thickness distributions. The Selig airfoils position the maximum thickness relatively close to the leading edge, whereas the CH 10-48-13 airfoil shifts the maximum thickness location further aft toward the mid-chord region. Of the three reference airfoils considered, the optimized outboard airfoil most closely resembles the CH 10-48-13 geometry, although it exhibits greater camber and a slightly more forward maximum thickness. The optimized inboard airfoil converged to a thicker airfoil profile than any of the reference airfoils, indicating that the optimization process favored additional thickness under the corresponding flow conditions. Overall, the similarity between the optimized geometries and established high lift airfoils provides a qualitative validation of the geometric parameterization and optimization framework.

B. Bayesian Optimization vs. NSGA-II

Figure 9 compares the convergence histories of the Bayesian Optimization (BO) and NSGA-II algorithms for the inboard and outboard optimization cases, providing insight into the convergence behavior, search efficiency, and overall performance of each optimization method.

The convergence histories reveal notable differences between the two optimization approaches. BO achieved rapid improvements during the early stages of the optimization process and reached higher final objective values for both cases. As discussed previously, BO constructs a surrogate model of the design space to balance exploration and exploitation. Consequently, the algorithm is able to efficiently identify promising regions of the design space and achieve large improvements in relatively few evaluations. This behavior is reflected in Fig.9, where significant jumps in objective value occur during early iterations before gradually approaching a plateau.

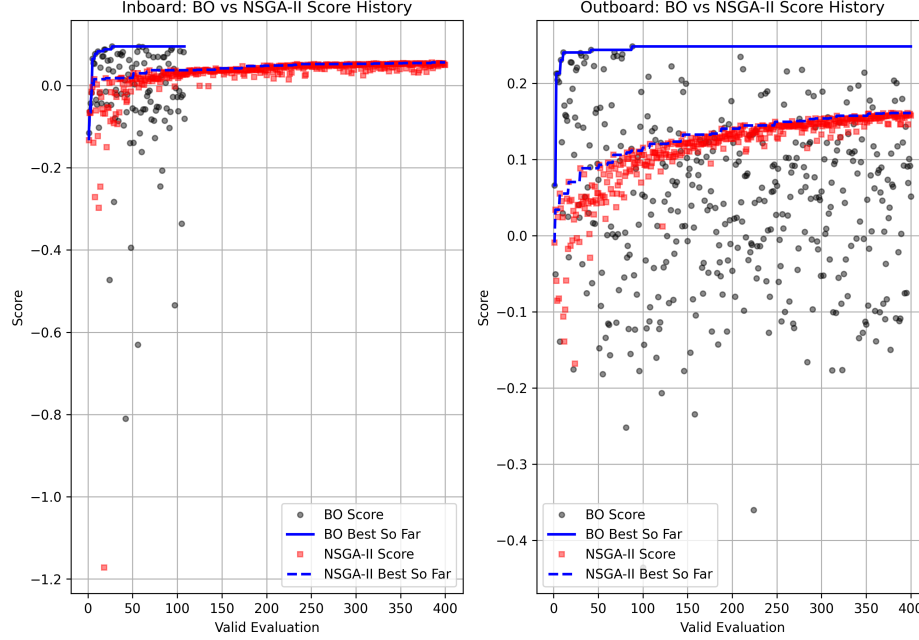


Fig. 9 Phase 1 BO and NSGA-II convergence history.

In contrast, NSGA-II exhibited a more gradual convergence pattern. NSGA-II evolves a population of candidate solutions through selection, crossover, and mutation. This strategy promotes broader exploration of the design space, but generally requires more evaluations to converge for low-dimensional optimization problems. As a result, the NSGA-II convergence history displays a smoother progression toward the final solution, with smaller incremental improvements occurring throughout the optimization process.

To further validate the results from the BO framework, the final geometries from both optimization strategies are compared in Fig.10.

Despite employing fundamentally different strategies, the optimized geometries produced by the BO and NSGA-II implementations exhibit many of the same overall characteristics. The outboard case exhibits larger geometric modifications than the inboard case, with both algorithms converging toward more aggressive high lift profiles. In comparison, the inboard optimization case produced smaller geometric changes relative to the seed airfoil. These trends are consistent with the aerodynamic objectives and operating conditions prescribed for each optimization case.

The similarity between the BO and NSGA-II results provides an additional level of validation for the implemented optimization strategy. The consistent improvement relative to the seed airfoil indicates that the geometric parameterization, aerodynamic analysis workflow, and objective function formulation are successfully guiding the optimization process toward favorable regions of the design space.

C. Limitations and Future Improvements

Several limitations and challenges were encountered during the development and implementation of the optimization framework. The most significant challenge was the integration of the MSES aerodynamic solver. While MSES provides relatively fast aerodynamic evaluations compared to higher-fidelity CFD methods, it was found to be highly sensitive to both geometric smoothness and flow conditions. Small geometric irregularities, aggressive parameter combinations, or unfavorable flow conditions frequently resulted in solver divergence. As a result, considerable effort was required to develop geometric constraints and validity checks capable of generating reasonable candidate airfoils.

The present study was also limited to two-dimensional aerodynamic analysis, and three-dimensional effects such as wingtip vortices, end-plate interactions, and spanwise flow effects were not considered. Consequently, the aerodynamic performance predicted in this study should only be used to inform design decisions rather than to predict real-world wing performance.

Future work includes the extension to the originally-proposed Phase 3, which would explore the ideal secondary element angle of attack α for low drag X-Mode. Due to MSES sensitivity, the α sweeps run were sporadic in which angles reached converged solutions. Thus, the scores for the phase were deemed illegitimate and more work is needed to determine the optimal X-Mode configuration. Integration with higher-fidelity CFD solvers would allow for more accurate aerodynamic evaluations of the geometry, and thus results could be reported. Finally, extension of the framework to

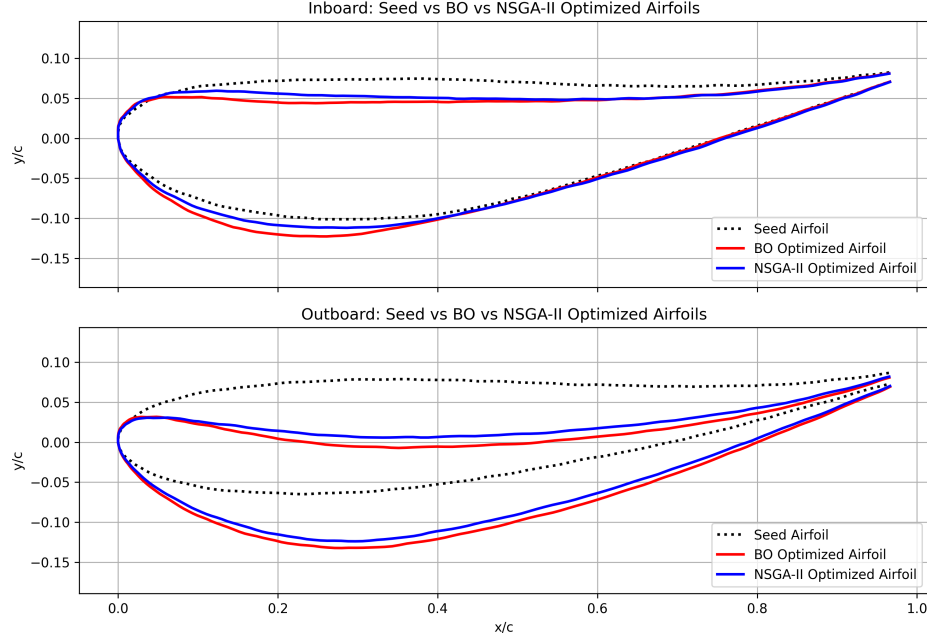


Fig. 10 Seed, BO, and NSGA-II Phase 1 airfoil comparison.

three-dimensional wing geometries would enable more realistic evaluation of aerodynamic performance and allow direct consideration of three-dimensional effects.

VII. Conclusion

This project presented a phased aerodynamic optimization framework for the design of a Formula 1 rear wing tailored to the Circuit de Monaco. The framework combined airfoil parameterization, geometric generation, viscous aerodynamic analysis using MSES, and Bayesian Optimization to identify improved aerodynamic configurations under representative racing conditions. By separating the optimization into multiple phases, the effects of main element geometry and secondary element geometry could be investigated independently while maintaining a manageable design space.

For the Phase 1 optimization studies, significant aerodynamic improvements were achieved for both the inboard and outboard operating conditions. The optimization consistently favored increased camber, resulting in substantial increases in downforce generation relative to the seed airfoils. Phase 2 demonstrated that further improvements could be obtained through optimization of the secondary element geometry, with highly cambered and relatively thin designs producing the highest performing configurations.

Validation studies using an independently implemented NSGA-II optimization framework produced similar optimized geometries and aerodynamic trends despite utilizing a fundamentally different optimization strategy. The agreement between the two approaches indicates that the observed improvements are representative of meaningful aerodynamic trends. Comparisons with established high lift low Reynolds number airfoils from the UIUC Airfoil Database further demonstrated that the optimized geometries exhibit characteristics commonly associated with successful aerodynamic designs.

Although the present work was limited to two-dimensional aerodynamic analysis and Phase 3 was not completed, the results demonstrate that Bayesian Optimization is an effective and computationally efficient approach for aerodynamic shape optimization problems involving expensive, gradient-free analyses. Future work will focus on completing the secondary element angle of attack optimization phase and extending the framework to higher-fidelity 3D simulations. Overall, the framework developed in this project establishes a foundation for simulation-driven aerodynamic optimization of motorsport wing systems.

VIII. Individual Contributions

Project responsibilities were initially divided according to major development areas. Emily developed the geometric parameterization framework and aerodynamic scoring functions. Anthony developed the MSES solver interface and aerodynamic analysis workflow. Mariah developed the optimization framework and implementation.

All team members contributed to the initial project architecture, framework design, and optimization methodology selection. In addition to their primary development responsibilities, all team members contributed to code integration, debugging, validation case development, testing, and refinement of the final framework. Report writing responsibilities were divided approximately equally among team members, with each author primarily responsible for documenting the components and results to which they contributed to most significantly during development.

Acknowledgments

The authors would like to thank Professor Kochenderfer and the course staff for an excellent quarter and for creating a course that made design optimization both approachable and genuinely enjoyable.

The authors would also like to thank Mark Drela for developing and making MSES available to us as graduate students, which served as the aerodynamic analysis tool used throughout this project. In addition, we are grateful to Matt Lauer for developing and maintaining the open source pymead framework and for providing the extensive documentation that made its use and customization possible.

Finally, the authors would like to apologize for the length of this report. What began as a relatively straightforward class project quickly turned into something we genuinely enjoyed working on, and each result seemed to lead to another question worth exploring. Before long, we found ourselves continuously adding new studies, figures, and analyses. While the report may be longer than originally intended, this project was one of the most enjoyable and rewarding experiences we have had during our time at Stanford.

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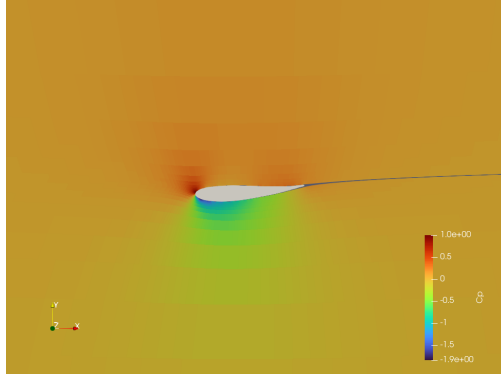
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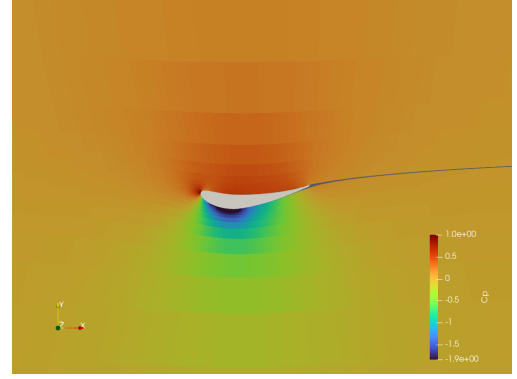
A. Additional Optimization Results

The appendix provides images with supplementary information about the optimization process for the two phases.

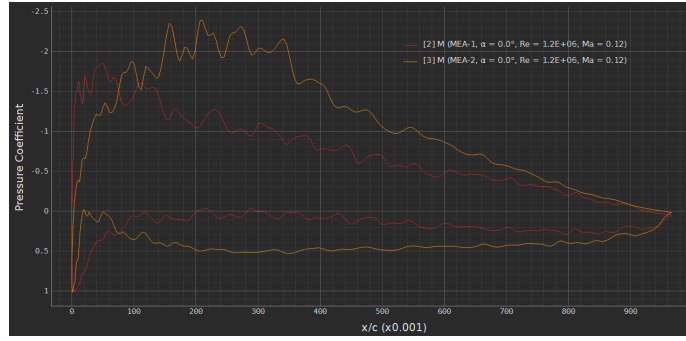
A. Phase 1: Outboard Optimization



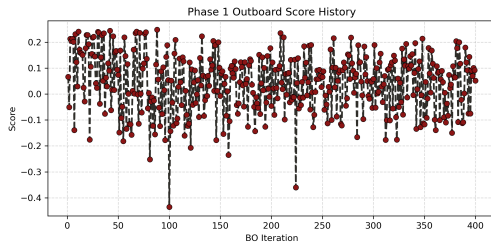
(a) Seed airfoil



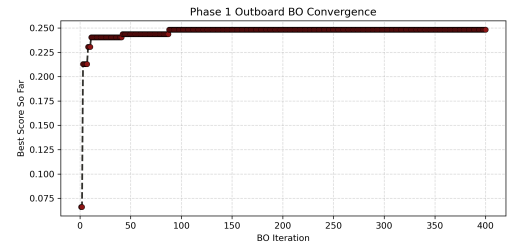
(b) Optimized airfoil



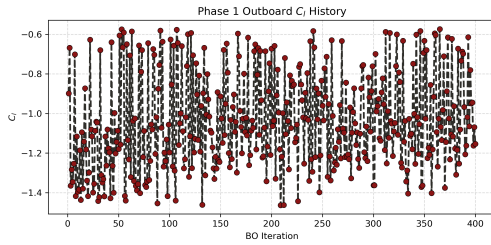
(c) C_p comparison



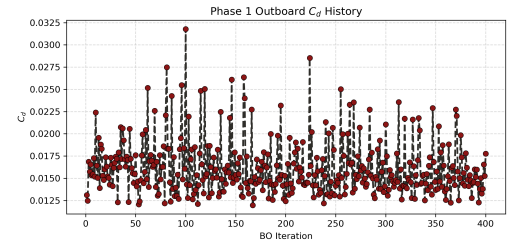
(d) Score history



(e) Best score convergence



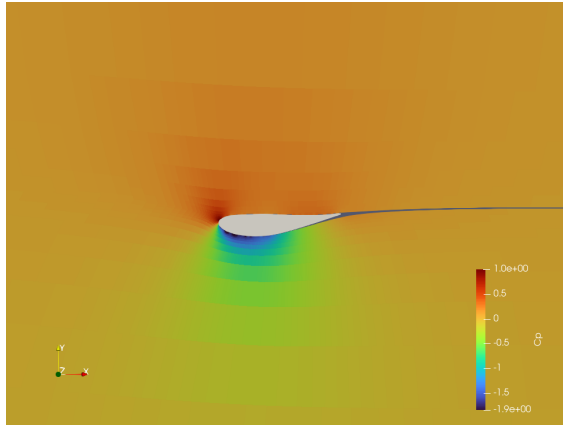
(f) c_l history



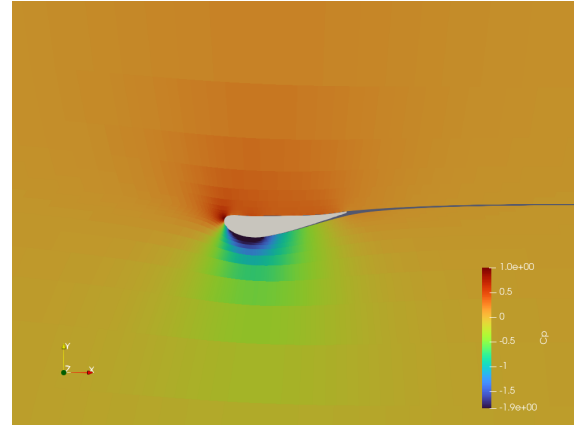
(g) c_d history

Fig. 11 Phase 1 outboard optimization results and Bayesian optimization history.

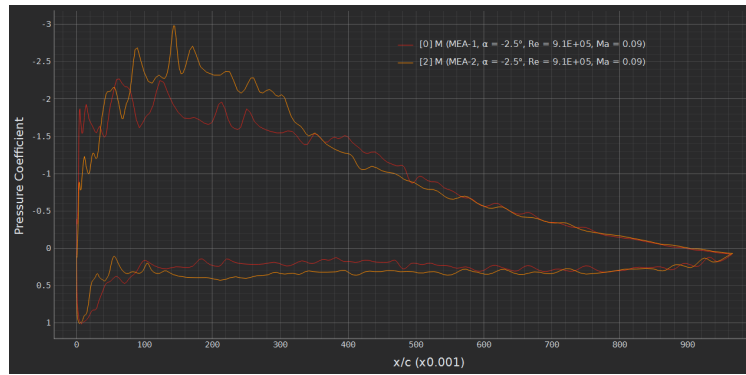
B. Phase 1: Inboard Optimization



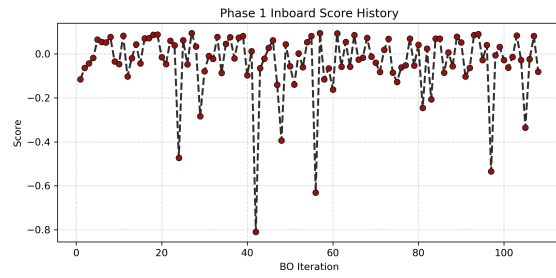
(a) Seed airfoil



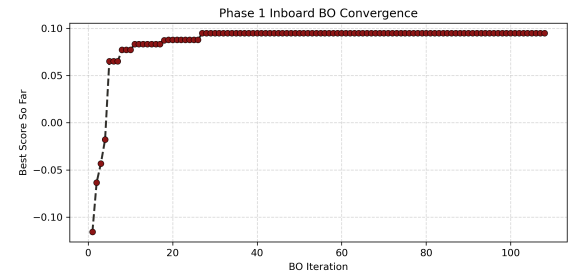
(b) Optimized airfoil



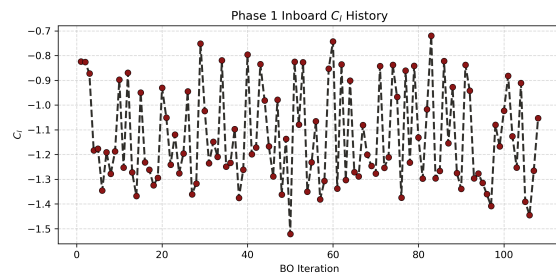
(c) C_p comparison



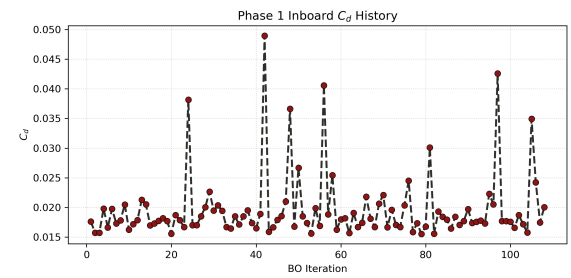
(d) Score history



(e) Best score convergence



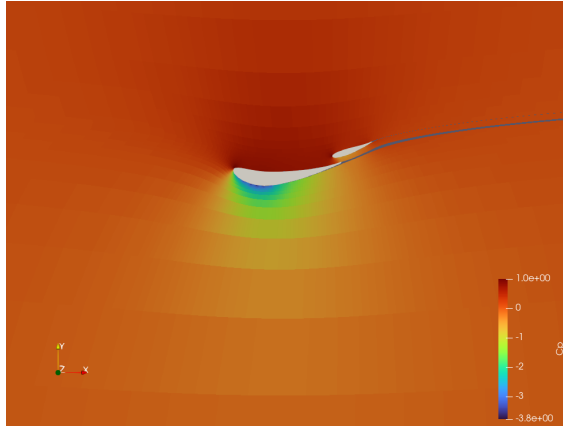
(f) C_l history



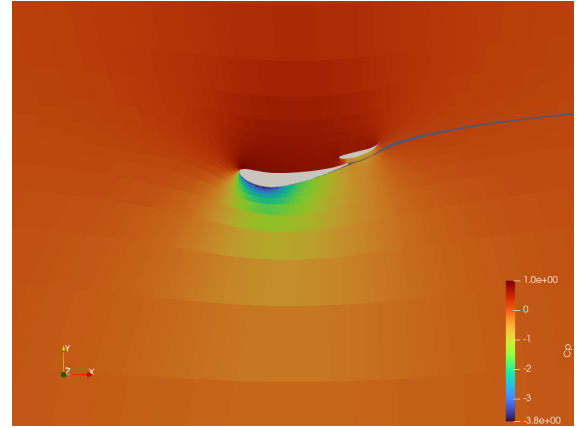
(g) C_d history

Fig. 12 Phase 1 inboard optimization results and Bayesian optimization history.

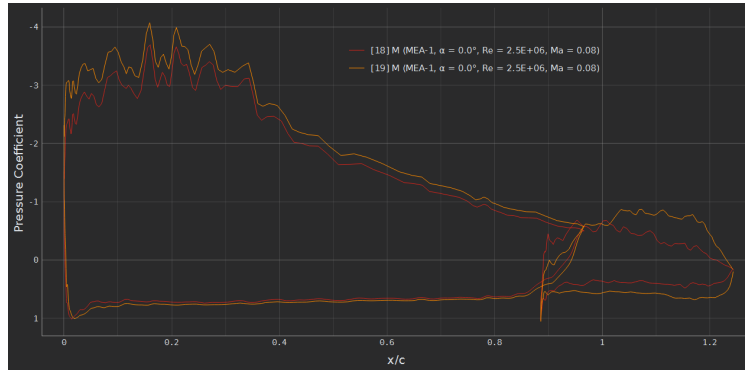
C. Phase 2: Outboard Optimization



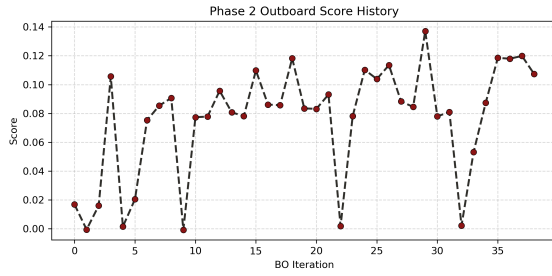
(a) Seed airfoil



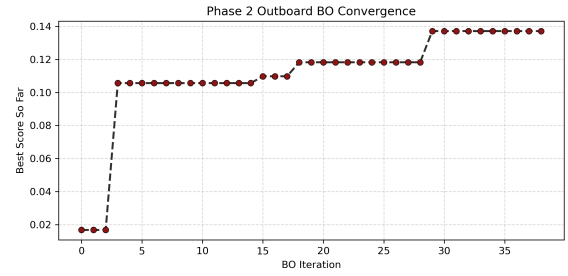
(b) Optimized airfoil



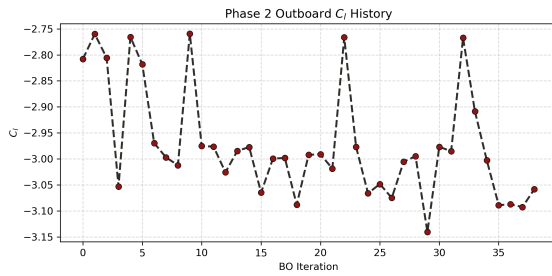
(c) C_p comparison



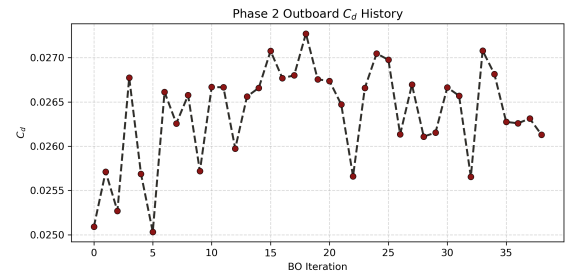
(d) Score history



(e) Best score convergence



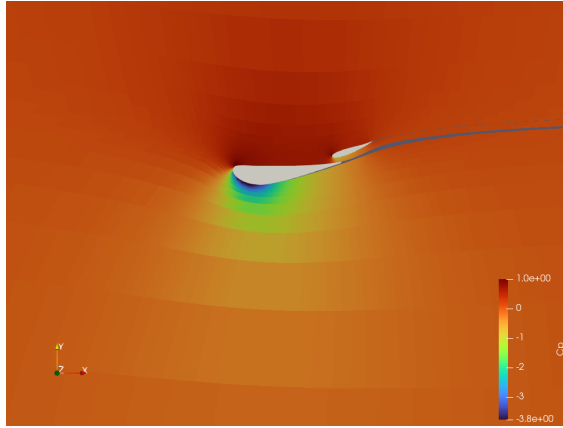
(f) c_l history



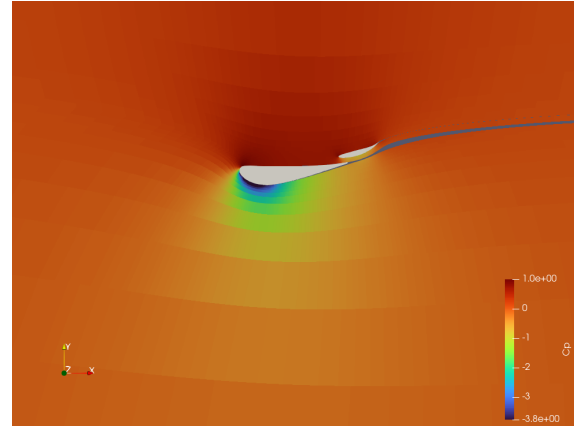
(g) c_d history

Fig. 13 Phase 2 outboard optimization results and Bayesian optimization history.

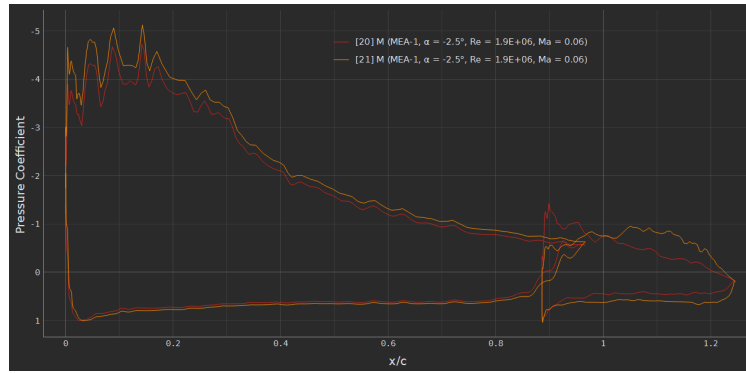
D. Phase 2: Inboard Optimization



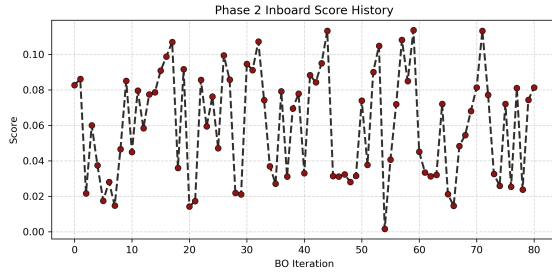
(a) Seed airfoil



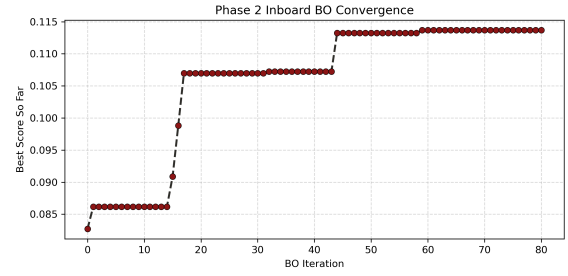
(b) Optimized airfoil



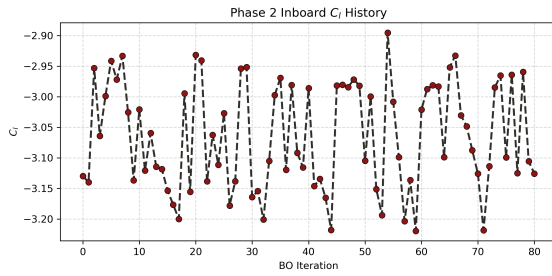
(c) C_p comparison



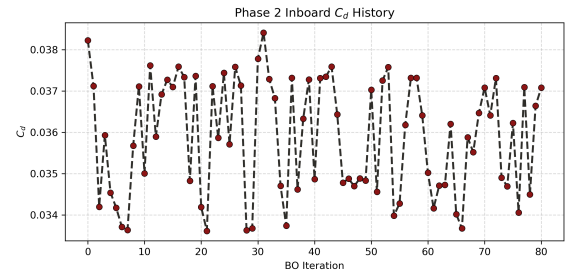
(d) Score history



(e) Best score convergence



(f) c_l history



(g) c_d history

Fig. 14 Phase 2 inboard optimization results and Bayesian optimization history.